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Exploratory Data Analysis of Large-Scale Industry Performance Dataset Using Univariate, Bivariate, and Multivariate Statistical Techniques

Introduction

Exploratory Data Analysis (EDA) is a statistical process used to understand datasets through summary statistics, visualizations, and relationship analysis before predictive modeling or machine learning.

This project performs comprehensive EDA on a large-scale industry performance dataset containing 15,000 records and 12 variables. The analysis focuses on identifying trends across industries, company performance metrics, employee distribution, revenue behavior, and market-related patterns.

Situation Overview

The primary objective of this project is to:

- Perform univariate, bivariate, and multivariate analysis
 - Detect and interpret outliers
 - Measure correlations between numerical variables
 - Analyze skewness and kurtosis
 - Extract meaningful business insights from industry data
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Dataset Overview

- Rows: 15,000
- Columns: 12
- Data Types: Numerical, Categorical, Date

Dataset Features

- id
- company_name
- industry

- country
- employee_count
- annual_revenue_million
- profit_margin_percent
- founded_year
- customer_count
- market_rating
- created_date
- region

Download the dataset “[industry_dataset.csv](#)” and upload the dataset into your project environment.

Steps in EDA for Industry Dataset

Univariate Analysis

Univariate analysis examines one variable at a time.

Numerical Variables

For each numerical column, the following statistical measures were computed:

- Mean
- Median
- Standard Deviation
- Minimum & Maximum
- Skewness
- Kurtosis

Example Insights

- annual_revenue_million showed positive skewness indicating that most companies generated moderate revenue while few companies generated extremely high revenue.
- employee_count exhibited high variance due to differences between small and large enterprises.
- profit_margin_percent showed relatively stable distribution across industries.
- customer_count displayed moderate right skew caused by large customer bases in retail and technology sectors.

Categorical Variables

Frequency distributions were analyzed for:

- industry
- country
- region

Example Findings

- Technology and Retail industries had the highest number of companies.
 - Majority of companies were located in North America and Asia.
 - Manufacturing companies showed relatively lower market ratings compared to Technology companies.
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Outlier Detection

Outliers were detected using:

- Boxplots
- Interquartile Range (IQR) Method
- Z-score Method

Observations

- Extreme values were present in annual_revenue_million.
 - High outliers in employee_count represented multinational enterprises.
 - Some companies showed exceptionally high customer_count, especially in the Retail sector.
 - Outliers were retained since they represent genuine large-scale organizations and important business scenarios.
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Bivariate Analysis

Bivariate analysis studies relationships between two variables.

Numerical vs Numerical

Correlation analysis was performed using Pearson Correlation.

Key Relationships

Strong Positive Correlation Between:

- employee_count and annual_revenue_million
- customer_count and annual_revenue_million

Moderate Correlation Between:

- market_rating and profit_margin_percent

Weak Correlation Between:

- founded_year and annual_revenue_million

Findings

- Companies with larger customer bases generally produced higher revenue.
 - Higher employee counts often corresponded to increased annual revenue.
 - Older companies did not always generate higher profits.
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Numerical vs Categorical

Examples:

- Industry vs average annual revenue
- Region vs employee count
- Country vs market rating

Findings

- Technology companies generated the highest average annual revenue.
 - Healthcare companies showed stable profit margins.
 - North American companies achieved slightly higher market ratings.
 - Manufacturing sector showed higher employee counts compared to Finance companies.
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Correlation Matrix Analysis

A correlation heatmap was generated for numerical variables.

Observations

annual_revenue_million strongly correlated with:

- employee_count
- customer_count

profit_margin_percent showed moderate correlation with:

- market_rating

founded_year had weak correlation with most operational variables.

No severe multicollinearity issues were identified among independent features.

Multivariate Analysis

Multivariate analysis examines interactions among three or more variables.

Examples

Industry × Revenue × Employee Count

→ Technology companies with larger employee bases generated the highest revenue.

Region × Market Rating × Profit Margin

→ North American companies showed better ratings and slightly higher profitability.

Country × Customer Count × Revenue

→ Retail companies in India and the USA handled significantly larger customer bases.

Industry × Founded Year × Revenue

→ Newly established Technology firms showed rapid revenue growth compared to older Manufacturing firms.

These patterns provide insights into operational efficiency, growth strategies, and industry performance.

Skewness and Kurtosis Interpretation

Skewness

Positive skewness was observed in:

- annual_revenue_million
- customer_count
- employee_count

This indicates that most companies operate at medium scale while a smaller number of enterprises dominate the market.

Kurtosis

High kurtosis was observed in:

- annual_revenue_million
- employee_count

This suggests the presence of extreme high-value organizations and large multinational companies.

Understanding skewness and kurtosis helped identify:

- Distribution asymmetry
- Extreme business behaviors
- Market concentration patterns
- Revenue dominance by large enterprises

Key Statistical Findings

- Revenue distribution was positively skewed.
- Large enterprises contributed significantly to total revenue.
- Technology sector dominated in revenue generation.
- Retail sector handled the highest customer volumes.
- Employee count strongly influenced company revenue.
- Outliers represented genuine large-scale companies and were retained for analysis.

Conclusion

This project successfully applied:

- Univariate statistical analysis
- Bivariate relationship analysis
- Multivariate interaction analysis
- Correlation study
- Outlier detection
- Skewness and kurtosis measurement

The analysis provides valuable insights for:

- Business intelligence
- Market trend analysis
- Industry benchmarking
- Revenue forecasting
- Strategic decision-making

Libraries Used

In Python, the following libraries were used during the EDA process:

- **NumPy** for numerical operations and multidimensional array handling.
- **Pandas** for data manipulation and statistical analysis.
- **Matplotlib** for data visualization and plotting.
- **Seaborn** for advanced statistical visualizations and heatmaps.

These libraries form the core foundation of Exploratory Data Analysis in Python.

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